W-Net: A Deep Network for Simultaneous Identification of Gulf Stream and Rings From Concurrent Satellite Images of Sea Surface Temperature and Height

Devyani Lambhate, Raghav Sharma, Jenifer Clark, Avijit Gangopadhyay, and Deepak Subramani, *Member, IEEE*

***Abstract*— Accurate digitization of synoptic ocean features is crucial for climate studies and the operational forecasting of ocean and coupled ocean–atmosphere systems. Today, for some North Atlantic operational regional models, skilled human experts visualize and extract the gulf stream and rings (warm and cold eddies) through an extensive and knowledge-based manual process. To automate this task, we develop a dynamics-inspired deep learning system that extracts the Gulf Stream and rings from concurrent satellite images of sea surface temperature (SST) and sea surface height (SSH). We pose the above problem as a multilabel semantic image segmentation problem. A novel deep convolutional neural network architecture named W-Net, with two parallel encoder–decoder branches, is developed to perform the segmentation. The W-Net's one branch is the SST branch (accepts SST image as input) and another is the SSH branch (accepts SSH as input), and the final output is a segmentation of gulf stream, warm rings, and cold rings. A dataset consisting of SST, SSH, and manual feature annotation (ground truth) from 2014 to 2018 is used for training. For gulf stream, we obtain 82.7% raw test accuracy and a low error of 4.39% in the detected path length. For the Rings, we obtain more than 71% raw eddy detection accuracy. A detailed ablation study and an examination of both SST and SSH parts of the network are presented to understand how the deep neural network learns to segment the gulf stream's meandering path and Rings accurately.**

***Index Terms*— Advanced Very High Resolution Radiometer (AVHRR), convolutional neural network (CNN), eddies, Gulf Stream, Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA), rings, semantic segmentation, western boundary currents.**

I. Introduction

AUTOMATED detection and accurate digitization of synoptic ocean features, such as boundary currents, eddies, and fronts, is a long-standing need for real-time operational oceanography, weather forecasting, and climate studies. Specifically, the Feature-Oriented Regional Modeling System (FORMS) for operational forecasting of the northwest Atlantic [1], [2] requires accurate digitization of the gulf stream and rings [3]. Several other scientific applications, including climatic regime shift studies and their implication to ecosystem [4]–[8], and ocean economic activities such as fisheries and shipping [9], also rely on the identification of gulf stream and rings.

Briefly, the gulf stream, a western boundary current driven by powerful winds, is a warm and swift ocean current that originates in the Gulf of Mexico, maneuvers around Florida's tip, follows the United States of America's eastern coastline, and finally flows toward the European continent. Among all western boundary currents, gulf stream is the largest and has the most influence on the climate. It is highly variable in both space and time, typically with a width of 100–150 km, depth of 800 m along the east coast that gradually increases to 3000–4000 m of depth in the open ocean (75° W–45° W), and surface currents of about 2.5 m/s. It transports significant water and heat with it and provides nutrients to northern waters, making the northwestern Atlantic biologically active and affecting northwestern Europe's temperature. As the gulf stream flows along the U.S.'s eastern coastline, sections of moving water swirl off from the gulf stream and loop back on themselves, forming whirlpools or eddies called gulf stream rings. These eddies appear on both sides of the gulf stream, creating a meandering loop cut off from the current. The eddies that break from the gulf stream and go into the Sargasso sea are cyclonic and typically have a cold temperature core. In contrast, those entering the shelf waters are anticyclonic and have a warm temperature core. The former is called a warm core ring (WCR) or warm eddy (WE), and the latter is called a cold core ring (CCR) or cold eddy (CE).

Characterizing and predicting the gulf stream and rings have been of interest to researchers since the early 1970s (see [10]). The most successful technique has been identifying and hand annotating by expert human analysts (see Jenifer Clark's Gulf Stream [11]) based on data from satellites, ships, and available *in situ* instruments. Forty years of such charts (over 5000 charts, 2–3 times a week) have been analyzed recently,

Manuscript received April 6, 2021; revised June 5, 2021; accepted July 6, 2021. Date of publication July 26, 2021; date of current version January 14, 2022. This work was supported in part by the Internal Grant of IISc toward hardware for deep learning (Startup Grant to DS), in part by the INSPIRE Faculty Fellowship from the Department of Science and Technology, Government of India under Grant IFA18-ENG271, and in part by the Arcot Ramachandran Young Investigator Award. The work of A. Gangopadhyay was supported by NSF under Grant OCE-1851242. *(Corresponding author: Deepak Subramani.)*

Devyani Lambhate, Raghav Sharma, and Deepak Subramani are with the Department of Computational and Data Sciences, Indian Institute of Science, Bengaluru, Karnataka 560012, India (e-mail: deepakns@iisc.ac.in).

Jenifer Clark is with Jenifer Clark's Gulfstream, Dunkirk, MD 20754 USA.

Avijit Gangopadhyay is with the School for Marine Science and Technology, University of Massachusetts Dartmouth, Dartmouth, MA 02747 USA.

Digital Object Identifier 10.1109/TGRS.2021.3096202

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revealing unique seasonal and inter-annual asymmetries in gulf stream ring formation [12]. The Warm Core Ring formation over the whole region (75°–55° W) increased from an average of 18 per year during regime 1 (1980–1999) to 33 per year during regime 2 (2000–2017) [7]. The Cold Core Ring data are described and analyzed in [13]. Semi-automated and automated methods from classical image processing and computer vision to detect these ocean features from satellite images started appearing in the early 1980s (see [14], [15]). Still, their capabilities are inadequate for modern oceanographic and climate study requirements.

Sea Surface Temperature (SST) and Sea Surface Height (SSH) are the primary oceanographic variables that characterize the gulf stream and rings. As such, we could use SST and SSH data for automated, simultaneous identification of these synoptic ocean features. The critical challenge is developing a robust and efficient algorithm that can accurately and simultaneously identify the gulf stream and its warm and cold rings and automatically produce a map with these synoptic features. Our goal in this article is to develop such an algorithm using deep learning.

In what follows, first, we define the problem statement and review the literature for related work from the fields of remote sensing, oceanography, medical image segmentation, and advanced image processing (Section I-A). Next, we describe our new deep W-Net architecture, training procedure, and evaluation metrics (Section II). Then, we show and analyze results from applying the network to the test dataset (Section III), followed by a discussion of the ablation studies and utility of the new W-Net (Section IV). Finally, we conclude with a summary and directions for future work (Section V).

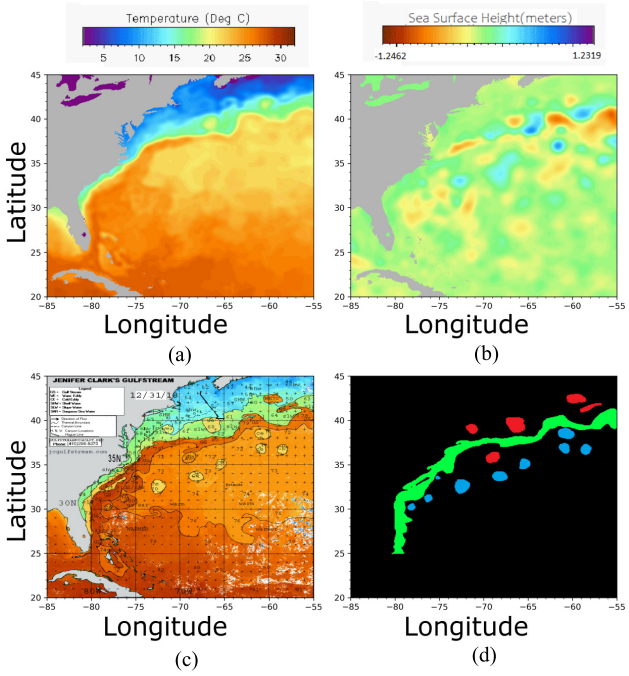


Fig. 1. Representative map of the region with gulf stream and rings. (a) SST Map, (b) SSH Map, (c) Jenifer Clark's hand annotations and (d) Ground Truth extracted from the hand annotations in the region between 85° W and 55° W, 20° N and 45° N on December 31, 2018.

A. Problem Statement and Related Work

We pose the task of identifying the gulf stream and rings as a problem of image segmentation, defined as follows (Fig. 1). At a particular time, we have concurrent images of the SST [Fig. 1(a)] and SSH [Fig. 1(b)], from which automatic annotations of the gulf stream, warm rings, and cold rings have to be made, as is done by a human [Fig. 1(c)]. Fig. 1(d) shows the manual segmentation that serves as the ground truth for our automated image segmentation task. Our problem statement is to develop a deep learning algorithm that automatically produces a segmentation as in Fig. 1(d).

Image segmentation, i.e., dense pixel-wise classification of an image into multiple class labels, is a widely studied fundamental problem in computer vision [16]. The segmentation methods can be broadly classified into pre- and post-deep learning eras. The most popular techniques include threshold-based approaches [17], Partial Differential Equation (PDE)-based level set approaches [18], [19], and specific medical imaging techniques [20] in the pre-deep learning era. The earliest ocean feature detection approach was to employ an edge detection algorithm to remotely sensed Sea Surface Temperature maps [21], [22]. Geometric pattern mapping and mathematical morphology [23] were also reported in the literature. Eddy detection from infrared satellite images by an iterative averaging of temperature gradients has also been proposed [24]. These automated methods had somewhat limited success in detecting the gulf stream north wall and warm eddies, but cold eddy detection was still problematic. In the current deep learning era, deep convolutional neural networks (deep-CNNs) have found great success for image segmentation tasks, including in remote sensing applications [25]–[30]. The fully convolutional network for image segmentation [31] and U-Net [32] developed for medical image segmentation have proved highly successful in several remote sensing applications [33]. U-Net introduced the powerful concept of encoder–decoder deep networks that work well with small annotated datasets, a feature common to both biomedical imaging and remote sensing. Here, the encoder network performs image contraction using max-pooling layers and the decoder network performs expansion using convolutional transpose layers. The U-Net network is generally trained using Intersection over Union (IOU; Jaccard index) or dice coefficient (F1-score) between the predicted and the ground truth segmentation. Deep residual U-Nets [34] have also been used to segment roads from aerial images, and residual pyramid nets have been used for eddy detection [35]. Automating the task of identifying the gulf stream north wall using a single-layer neural network for classification and a dynamic-based approach has been reported [3]. In their approach, longitude, latitude, temperature, day, year, speed, and sea surface height anomaly were used as input to a single hidden layer neural network to classify this location as gulf stream north wall or not. Also, simultaneous detection of eddies was not performed, and the single-layer neural net's

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success in detecting GSNW was limited. The simultaneous detection problem is particularly challenging due to the large amplitude meandering of the gulf stream that obfuscates it with Rings downstream (northeast part of our domain) [36]. Treating the problem as one of image segmentation and applying deep learning to predict the segmentation is promising in this context. However, successful application of deep networks for performing ocean feature extraction as a semantic segmentation problem has not been reported in the literature so far, and this article aims to fill this gap.

B. Dataset Description

Our dataset consists of Sea Surface Temperature (SST) maps [Fig.1(a)], Sea Surface Height (SSH) maps [Fig.1(b)], and the manual annotations of gulf stream and eddies by an expert [Fig.1(c)]. Our focus area is the region bounded by 85°W to 55°W and 20°N to 45°N, covering approximately 9 million kilometer square of surface area.

1) Sea Surface Temperature Data: We use the Level 4 Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA) product [37] as our SST input. This product is produced daily by the Group for High-Resolution Sea Surface Temperature (GHRSST) at the U.K. Met Office using optimal interpolation. The analysis uses satellite data from the Advanced Very High Resolution Radiometer (AVHRR), the Spinning Enhanced Visible and Infrared Imager (SEVIRI), the Advanced Along Track Scanning Radiometer (AATSR), the Tropical Rainfall Measuring Mission Microwave Imager (TMI), the Advanced Microwave Scanning Radiometer-EOS (AMSRE), and *in situ* data from drifting and moored buoys. The data created in this analysis have a smooth SST field and was specifically produced to support SST data assimilation into Numerical Weather Prediction (NWP) models [37]. These data are available on a global 0.054° grid, i.e., the corresponding SST images have a resolution of approximately 5.5 km per pixel.

2) Sea Surface Height Anomaly Data: We use the gridded product of sea-level anomaly provided by copernicus marine environmental monitoring service (CMEMS [38]) as our SSH input. This product is constructed from multisatellite altimetric missions and is available on a global 0.25° grid, i.e., the corresponding SSH images have a resolution of approximately 27 km/pixel.

3) Jenifer Clark's Gulf Stream and Ring Annotation: We use the annotated maps in our area of interest produced by Jenifer Clark [11] based on an expert analysis of the seven- and three-day SST composite products from JHU [39], original 6-h AVHRR SST swaths, SSH maps, and other sources of *in situ* data (e.g., local fishermen). Fig. 1(c) shows an example of the annotated gulf stream and rings. We manually converted these annotations to a segmentation map of "gulf stream," "warm eddies," "cold eddies," and "background" as ground truth [Fig. 1(d)] for our supervised deep learning algorithm.

For the present application, we use SST, SSH, and the corresponding ground truth annotation data between 2014 and 2018. Three- or seven-day composite concurrent images are used as inputs to the deep network from the SST and SSH products to match the corresponding ground truth Jenifer Clark annotation. Overall, during this time, we had access to a total of 358 annotated images.

4) Challenges: One fundamental challenge of applying machine learning techniques to remote sensing data is that the number of labeled training data available is limited, making it difficult to train deep networks. We overcome this challenge by using data augmentation during training as described in Section II-B.

II. Deep Learning Model Development

A. W-Net Architecture

We propose a novel deep-CNN architecture that we named W-Net, based on the popular U-Net architecture [32]. Our architecture's key novelty (Fig. 2) is that it has two U-Net-like fully convolutional encoder–decoder branches. The W-Net with number of channels, layers, and a visualization of the different operations is shown in Fig. 2. One branch has sea surface temperature as input, and the other has sea surface height anomaly as input. Each downconvolution block has two unstrided convolution layers and a maxpool layer with a stride of two. Each upconvolutional block comprises a strided convolutional transpose layer and two convolutional layers. The central block that connects the encoder with the decoder is composed of only two unstrided convolutional layers. A ReLU activation is used after all the convolution layers in the network. Connecting each downconvolution and upconvolution layer of the same size is a skip connection [32] marked as a "COPY" operation in Fig. 2. This skip connection provides the output of the downconvolution not only to its immediate successor but also to upconvolutional layers further down the W-net, thereby allowing gradients to flow better aiding convergence [40]. After several experiments to tune the hyperparameters, we fixed the number of upconvolutional decoder blocks and downconvolutional encoder blocks in both SST and SSH branches to six. The decoder part of both the U-Net-like branches produces an intermediate four-channel output corresponding to the gulf stream, warm rings, cold rings, and background. These intermediate layer outputs are then combined channel-wise using a 1×1 convolutional filter to obtain a four-channel output. Then, a softmax layer is applied and the final segmentation map is produced as a one-hot representation.

As the SSH images are of lower resolution than SST (Section I-B), they are first passed through an additional upsample layer that performs bilinear interpolation. Bilinear interpolation was chosen after several studies of different upsampling strategies to identify the option that gave the best validation accuracy of the final segmentation.

B. Training

The W-Net model is implemented using the PyTorch library [41]. We used two Quadro RTX 8000 GPUs for training the W-Net model, each with 48-GB memory. The network is trained for 200 epochs, and the model with the best validation loss was saved (Fig. 3). Careful hyperparameter tuning was completed (not shown here; see [42], [43] for sensitivity to

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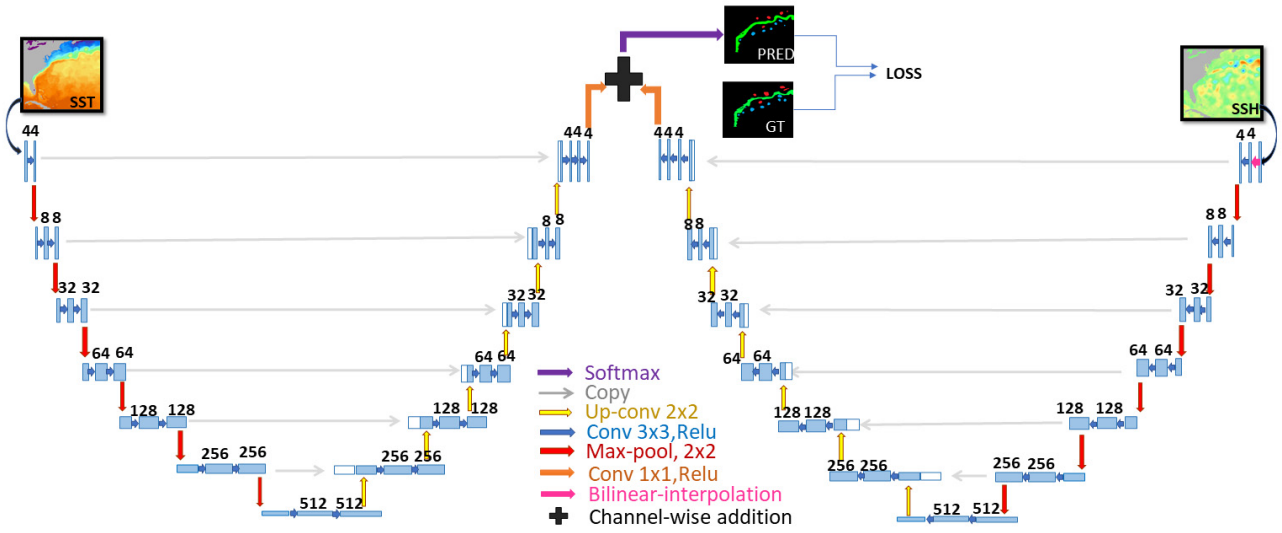


Fig. 2. W-Net Architecture: different components of the network's encoder and decoder layers marked with the number of channels, layers, and operations performed. The loss is backpropagated to both the SST and SSH branches during training.

network size), and the best combination that led to the lowest validation error (epoch 55) is chosen as the final model. Briefly, increasing the number of downconvolution beyond six reduced evaluation metrics, using more than two convolutional layers per block did not improve accuracy any further. Testing with unseen data is then performed on the saved model.

1) Train–Test Split: To ensure a representative train and test set, we form a test set with images of all weeks in a year but sampled from different years. The remaining data form the train set. From the train set, a representative validation set with 10% of the images is formed.

2) Data Augmentation: Because of the scarcity of labeled images, we perform augmentation by generating new images formed by rotating the SST, SSH, and ground truth images every 45°. In this procedure, every labeled data point gives seven new rotated images at different orientations. Further data augmentation did not increase validation accuracy.

3) Loss Function: The cross-entropy loss function is used for training the W-Net

$$\text{Cross-Entropy} = \sum_{i=1}^{N} -y_i \log \hat{y}_i \tag{1}$$

where y_i and $\hat{y}_i$ are the ground truth and prediction for the ith pixel, respectively, and N is the total number of pixels in the image. During the ablation study (Section IV-A), we also trained with other losses such as dice loss and a mix loss

$$\text{Dice-Loss} = 1 - \text{Dice Coefficient} \tag{2}$$

$$\text{Dice Coefficient} = \frac{2 \times (Y \cap \hat{Y})}{Y \cup \hat{Y} + Y \cap \hat{Y}} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FN} + \text{FP}} \tag{3}$$

$$\text{Mix-Loss} = \lambda\, \text{Dice-Loss} + (1 - \lambda)\, \text{Cross-Entropy} \tag{4}$$

where $\hat{Y}$ is the predicted output, Y is the true output, TP is the true positive, FN is the false negative, FP is the false positive, and λ is the mixing ratio.

The loss is backpropagated through both branches, and weights are updated by the Adam optimizer with a learning rate of 0.001. We also use skip connections and a 20% dropout regularization. The weights are initialized using Xavier initialization [44].

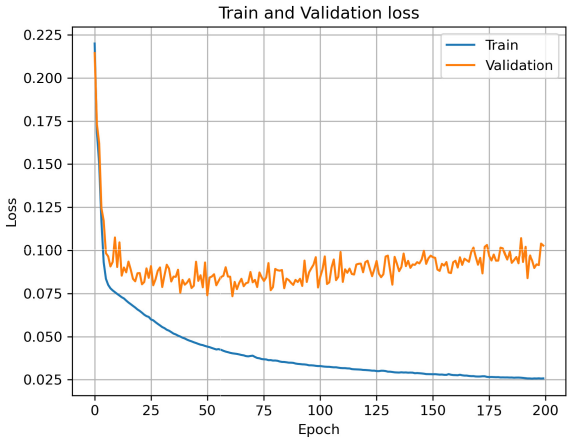


Fig. 3. Train and validation loss of final W-Net model across 200 training epochs.

C. Evaluation Metrics

Commonly image segmentation is done on natural images, for which IOU and dice coefficient are used for both evaluation and training [45]. However, in our application, we are interested in the dynamics of gulf stream and rings. As such, different dynamic-inspired metrics are necessary to evaluate the performance of the deep network. We devise metrics that compare the size, centroid, and count between the ground truth and the network prediction for eddies. For the gulf stream, we evaluate the path length difference and the difference between the ground truth and predicted Gulf Stream's centerline using the Hausdorff distance between those two centerlines. The different evaluation metrics and how they are computed are detailed next.

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1) Dice Coefficient/F1-Score: The dice coefficient is defined as twice the overlap area of the prediction and ground truth divided by the sum of areas of the prediction and ground truth. Mathematically, (3) defines the dice coefficient. The dice coefficient is the F1-Score and a single number that quantifies a network's performance [46]. It ranges from 0 to 1, where 0 indicates no similarity between predicted and ground truth and 1 indicates perfect similarity. This metric is positively correlated with the IOU (used in the original U-Net [32]), which is simply the ratio of overlap (intersection) and the union of predicted and ground truth. The Dice coefficient tends to measure the average performance, whereas IOU accuracy tends to measure the worst performance.

2) Fraction of Eddies Correctly Detected: The fraction is calculated as the ratio of the number of eddies correctly detected by the W-Net and the number present in the ground truth, that is,

$$\text{Fraction of Eddies Correctly Detected} = \frac{\text{No. of eddies correctly detected by W-Net}}{\text{No. of eddies present in the ground truth}}. \quad (5)$$

If there is any overlap between the predicted and ground truth eddies, the particular ground truth eddy is declared as detected. This metric is separately calculated for warm and cold eddies, giving us a good insight into our W-Net's performance in detecting gulf stream rings.

3) Eddy Centroid Difference: For all eddies declared as correctly detected, we compute the distance between the detected eddy's centroid and the corresponding ground truth eddy's centroid. Per image, we calculate the mean absolute centroid difference to get a metric that quantifies the average spatial location error in detecting eddies, that is,

$$\text{Eddy Centroid Difference} = ||\mathbf{r}_{c,\text{W}-\text{Net}} - \mathbf{r}_{c,\text{GT}}||_2 \quad (6)$$

where $\mathbf{r}_{c,\text{W}-\text{Net}}$ and $\mathbf{r}_{c,\text{GT}}$ are the coordinates of the centroid detected by the W-net and corresponding ground truth, respectively. Similar to the fraction of eddies detected, this metric is also computed separately for warm and cold eddies. The connected components library of OpenCV [47] is employed to calculate the centroid.

4) Eddy Area and Equivalent Diameter Difference: The areas of the detected and ground truth eddies are calculated using the number of pixels present in the predicted and ground truth, respectively. The equivalent diameter of a circle with the same area is calculated as $d_{\text{eq}} = 2((\text{area}/\pi))^{1/2}$. As before, per image, the mean absolute difference between the area and the equivalent diameter is calculated, that is,

$$\text{Eddy Area Difference} = |A_{\text{W}-\text{Net}} - A_{\text{GT}}| \quad (7)$$

$$\text{Equiv. Dia. Difference} = |d_{\text{eq,W}-\text{Net}} - d_{\text{eq,GT}}| \quad (8)$$

where $A_{\text{W}-\text{Net}}$ and A_{GT} are the areas of the eddy detected by the W-net and corresponding ground truth, respectively. We also divide the absolute equivalent diameter difference and the ground truth's equivalent diameter to arrive at a relative error metric, that is,

$$\text{Relative Diameter Difference} = \frac{|d_{\text{eq,W}-\text{Net}} - d_{\text{eq,GT}}|}{d_{\text{eq,GT}}}. \quad (9)$$

This ratio's mean provides an average relative error estimate. These metrics are calculated separately for the warm and cold eddies, and they collectively quantify the performance of the W-Net for correctly identifying the size of eddies.

5) Similarity of Gulf Stream Centerline: One key challenge in automated detection of the Gulf Stream is to capture its meandering path accurately. To evaluate our W-Net's performance in detecting the meanders, we compute the Hausdorff distance between the centerline of the detected and ground truth Gulf Stream. The Hausdorff distance is the supremum of all the distances from a point in one curve to the other curve's closest point. A Sobel filter [48] is applied to the segmentation map to identify the north and south lines of the gulf stream. A thinning operation is performed on the gulf stream segments to identify the centerline. We also calculate the mean and median of all the distances from a point in one curve to the other curve's closest point. Mathematically, the metrics are

$$\text{MCD} = \text{mean}\left\{ \sup_{\mathbf{x}\in\mathbf{G}_W} d(x, G_{\text{GT}}),\ \sup_{\mathbf{y}\in\mathbf{G}_{\text{GT}}} d(y, G_W) \right\} \quad (10)$$

$$\text{Med.} = \text{median}\left\{ \sup_{\mathbf{x}\in\mathbf{G}_W} d(x, G_{\text{GT}}),\ \sup_{\mathbf{y}\in\mathbf{G}_{\text{GT}}} d(y, G_W) \right\} \quad (11)$$

$$\text{Haus.} = \max\left\{ \sup_{\mathbf{x}\in\mathbf{G}_W} d(x, G_{\text{GT}}),\ \sup_{\mathbf{y}\in\mathbf{G}_{\text{GT}}} d(y, G_W) \right\} \quad (12)$$

where MCD is the mean curve distance, med. is median curve distance, Haus. is the Hausdorff distance, $d(p, Q)$ is the distance of the point p from the set Q, G_{GT} is the set of Ground Truth Gulf Stream's centerline, G_W is the set of W-net predicted Gulf Stream's centerline. Together, these three metrics give an excellent evaluation of the network's performance for detecting the Gulf Stream path.

6) Path Length Difference of the Gulf Stream: We also compute the absolute difference of the detected and ground truth Gulf Stream's centerline path length, that is,

$$\text{Path Length Difference} = |l_{G,\text{W}-\text{Net}} - l_{G,\text{GT}}| \quad (13)$$

where $l_{G,\text{W}-\text{Net}}$ is the length of the W-Net detected Gulf Stream's centerline and $l_{G,\text{GT}}$ is the ground truth Gulf Stream's centerline. We also estimate the ratio of path length difference to the length of the ground truth Gulf Stream to arrive at a relative error metric, that is,

$$\text{Relative Path Length Difference} = \frac{|l_{G,\text{W}-\text{Net}} - l_{G,\text{GT}}|}{l_{G,\text{GT}}}. \quad (14)$$

Together with the curve similarity metrics, these metrics estimate the W-Net's performance for detecting the complex meanders of the gulf stream.

III. Gulf Stream and Rings Detection With W-Net

The trained W-Net with the best network weights identified from the multiple training runs and ablation study is used to perform testing. The training and validation loss function (cross-entropy) with the number of epochs trained for the final W-Net is shown in Fig. 3. The weights that gave the minimum validation error are chosen.

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The different performance metrics (Section II-C) for ring (eddy) and gulf stream detection are shown in Table I. The average Gulf Stream detection accuracy is very high. We accurately detect the meanders, which has hitherto been unsuccessful using other automated techniques [3], [22]. The Gulf Stream accuracy metrics (Section II-C) in Table I show the quantitative comparison between the network prediction and the ground truth. The low values of the mean, median, and max of the distance between the W-Net prediction and ground truth centerlines show how well our W-Net predicts the meandering Gulf Stream. The low values of the mean and median path length difference indicate that our W-Net can predict the entire path length from south of Florida to the east of the domain. Furthermore, the path length difference ratio to the ground truth Gulf Stream's length in the ground truth indicates excellent performance with an error of only 4.39%.

On average, we correctly detect 76.25% of warm eddies and 71.47% of cold eddies. Comparing the detected and missed eddies' average equivalent diameter indicates that smaller eddies are missed on average. Thus, the availability of finer resolution SSH data should improve the accuracy of automated eddy detection. On average, the distance between the centroid of eddies that are correctly detected and their corresponding ground truth is 32.6 km (5.9 pixels) for the warm eddies and 21.15 km (3.85 pixels) for the cold eddies. To quantify the error in the detected eddies' size, we evaluate their areas, equivalent diameter, and the ratio of the error in equivalent diameter of correctly classified eddies to the average equivalent diameter of the correctly classified ground truth eddy (Table I). Notably, we see that the equivalent diameter difference is only 16.5% of the ground truth eddy's equivalent diameter for warm eddies. For cold eddies, this ratio is 17.1%. These numbers indicate that the size error is low in eddies that are detected accurately.

The predicted and ground truth segmentation of four representative images from the test set are shown in Fig. 4. These images are chosen from January, April, July, and October from different years to convey the network's average performance across seasons and years. The SST and SSH are inputs to the W-Net, and the hollow shape segmentation is predicted from the W-Net. The solid shape segmentation is the ground truth. Qualitatively, we see that the network predicts a segmentation that is visually similar to the ground truth. The Gulf Stream is accurately detected with low errors (mean curve distance) in all the four weeks shown. The eddy detection accuracy is very high for July 8, 2016, October 20, 2017, and April 30, 2018. On January 5, 2015, only five out of the ten cold eddies are correctly predicted. However, this week has extremely high accuracy for Gulf Stream and warm eddies. Note that other weeks from November to March have a higher fraction of detection, and some weeks in the other months have lower detection accuracy. Overall, our network's average performance is reasonably high, as indicated by the performance metrics in Table I.

For some weeks (e.g., January 5, 2015), the W-Net predicts additional eddies that are not present in the ground truth. Further analysis of all ground truth images to identify whether these eddies are missed in the human segmentation may be completed in the future.

TABLE I
W-NET PERFORMANCE METRICS

A. Gulf Stream Detection Performance		
Metric	***Value***	
Mean curve distance	23.65 km	
Median curve distance	21.12 km	
Max curve distance (Hausdorff distance)	145.09 km	
Mean path-length difference	169.4 km	
Median path-length difference	113.85 km	
Ratio of path-length difference to the length of the ground truth Gulf Stream	0.0439	
B. Eddy (Ring) Detection Performance		
Metric	***Warm eddies***	***Cold eddies***
Fraction of eddies correctly detected	0.7625	0.7147
Fraction of eddies correctly detected with diameter $\geq$ 75 km)	0.7936	0.8047
Eddy centroid distance	32.6 km	21.15 km
Eddy area absolute difference	7068.215 km^2	3166.268 km^2
Eddy equivalent diameter absolute difference	27.28 km	23.375 km
Ratio of eddy equivalent diameter absolute difference to the eddy equivalent diameter of the ground truth	0.165	0.171
Intersection over union	0.518	0.467
C. Size Analysis of Detected and Missed Eddies (Rings)		
Metric	***Warm eddies***	***Cold eddies***
Average diameter of detected eddies	76 km	58 km
Average diameter of missed eddies	53 km	46 km

IV. DISCUSSION

A. Ablation Study

To arrive at the final W-Net presented above and examine each network part's contribution, we performed a detailed ablation study. Each of the following model architectures was trained and tested.

1) *U-NET-SST:* U-Net-SST is a simple U-NET with three input channels corresponding to the RGB channels of the SST images. It has six encoder blocks and six decoder blocks.
2) *U-NET-SSH:* U-Net-SSH is a simple U-NET with three input channels corresponding to the RGB channels of the SSH images. It has six encoder blocks and six decoder blocks.
3) *U-NET-6:* U-Net-6 is a straightforward, simple U-Net with six input channels. The first three channels are RGB channels corresponding to the SST image, and the following three channels are the RGB channels corresponding to the SSH image.
4) *W-NET:* The W-Net is composed of two U-Net-like branches and is described in Section II-A. The cross-entropy loss function is used for training.
5) *W-Net-DICE:* W-Net-Dice's architecture is the same as W-Net, but while training, only Dice loss is used.

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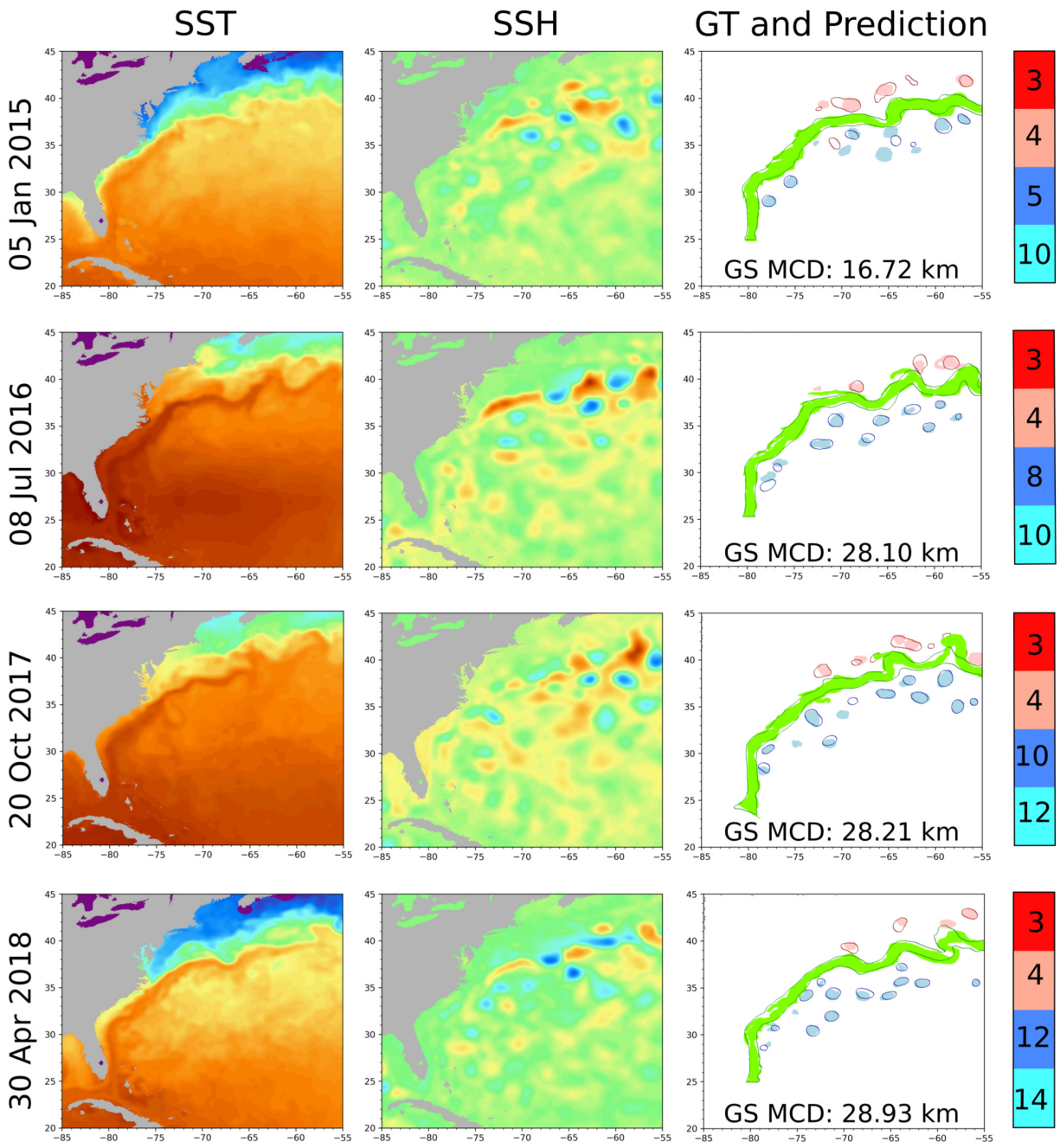


Fig. 4. W-Net Predictions on Test Set: SST, SSH inputs in the left and middle columns, W-Net's segmentation (hollow shapes) along with the ground truth segments (solid shapes) in the right column. Green color is for the Gulf Stream, blue is for the cold eddies, and red is for the warm eddies. The number of warm (cold) eddies correctly predicted and those present in the ground truth are enumerated in the darker red (darker blue) box and lighter red (lighter blue) box, respectively. The Gulf Stream Mean Centerline Distance (GS MCD) is also indicated in the right column. The four images chosen here are representative of the season and years and are an accurate depiction of the network's average performance. The color axis is the same as in Fig. 1.

6) *W-NET-MIX:* W-Net-Mix's architecture is also the same as W-Net, but while training, an equal combination of Dice and Binary Cross-Entropy loss is used.
7) *W-NET-LOW-SSH:* W-Net-Low-SSH's architecture is similar to W-Net, but SSH input is at its original resolution on which convolutions are applied. At the end, a nearest neighbor interpolation is performed to get the final segmentation.
8) *RES-W-NET:* In RES-W-Net, we used residual connections inside all the encoder networks' convolutional blocks (both SST and SSH). Training is done similar to the W-NET.
9) *Y-NET:* Y-Net model has two encoders: SST encoder and SSH encoder but a single decoder. The outputs from the two encoders are combined and given to the decoder to get the final segmentation.

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10) *W-NET-Summer:* Same architecture as W-Net, but only summer (May-Sep) month images are used during training and testing.
11) *W-NET-Winter:* Same architecture as W-Net, but only winter (Oct-Apr) month images are used during training and testing.

The dice coefficient for all the different networks is reported in Table II. Our model development first started with a vanilla U-Net with only SST (U-Net-SST) as input and only SSH (U-Net-SSH) as input. These networks had a three-channel input comprising RGB channels of SST and SSH. U-Net-SST reported poor accuracy numbers, whereas U-Net-SSH achieved better performance. Thus, we understand that SSH has more predictive power than SST for simultaneous detection of Gulf Stream and Rings. Next, we trained U-Net-6 with a six-channel input comprising RGB channels of both SST and SSH. This network improved the eddy detection accuracy further with only a marginal improvement in the Gulf Stream Dice coefficient. This observation indicates that SST data augment SSH in accurately identifying eddy sizes. Next, we developed the W-Net architecture that is this article's main contribution (Section II-A). The W-Net improved cold eddy detection dice coefficient by 6% points and warm eddy detection Dice coefficient by approximately 2% points. All the above networks were trained with cross-entropy loss. After that, we made efforts to improve the W-Net accuracy by trying dice loss and a mixture of cross-entropy and dice loss. To understand how the initial SSH interpolation affects the W-Net's accuracy, we trained another architecture W-Net-Low-SSH in which the SSH input is used at its original resolution (25 km) to which convolutions are applied. In this network, just before the channel-wise addition, a nearest neighbor interpolation is applied. This network performs worse than the W-Net for detecting eddies but gives comparable accuracy for gulf stream, suggesting that higher resolution input helps in detecting eddies more accurately. We also tried a Res-W-Net architecture where residual connections [49] were added to the U-Net-like branches of W-Net. A different architecture (christened as Y-Net) with two encoders but only one decoder was also implemented and trained. Among all the architectures (Table II), and the W-Net trained with the cross-entropy loss function produced the best dice coefficient metrics. The Dice coefficient metric was positively correlated with the dynamic-inspired performance metrics reported for the final W-Net in Section III. As such, analyzing the dice coefficient is sufficient to evaluate the performance of the different architectures. Note that several other loss functions, including IOU, were tried but not reported here as they did not improve the predictions. Even though dice coefficient and IOU metrics are great to compare performance, we observe that they are not good loss functions for training in our application. We hypothesize that the reason could be because cross-entropy has well-behaved gradients than the other metrics. Especially with a relatively small dataset (compared to other segmentation applications), the training algorithm finds a better set of network weights when cross-entropy is used in our application. Note that for other applications, results have been reported where training with dice coefficient has produced a better result than training with categorical cross-entropy loss [50], but the dataset is much bigger than what we use. Finally, to understand the seasonality in detection accuracy, we trained and tested the W-Net only with summer month and winter month images in W-Net-Summer and W-Net-Winter networks. The W-Net-Summer performs worse than W-Net-Winter, especially for detecting cold eddies in the Sargasso sea. This result is expected as gradients in SST in the summer months are not as prominent as the winter months; therefore, the network gets little help in learning weights accurately when trained only with seasonal data. Further breakdown into fall, winter, spring, and summer can be done with more years of data in the future.

TABLE II
DICE COEFFICIENT OF DIFFERENT EXPERIMENTAL MODELS IN ABLATION STUDY

	Dice Coefficient		
	Cold Eddies	***Warm Eddies***	***Gulf Stream***
U-NET-SST	0.018	0.037	0.207
U-NET-SSH	0.495	0.402	0.786
U-NET-6	0.635	0.598	**0.828**
W-NET	**0.682**	**0.637**	**0.827**
W-NET-DICE	0.40	0.51	0.70
W-NET-MIX	0.617	0.592	**0.827**
W-NET-LOW-SSH	0.501	0.517	0.812
RES-W-NET	0.02	0.10	0.33
Y-NET	0.453	0.538	0.806
W-NET-Summer	0.054	0.306	0.695
W-NET-Winter	0.467	0.617	0.806

Recurrent neural network (RNN) architectures are also useful for the problem at hand. In fact, there are several new use cases and configurations possible with RNNs. As such, we postpone the inclusion of RNN to a future manuscript and focus only on the simultaneous identification problem with concurrent images as input.

B. Insight Into the Neural Learning System

To elucidate the contribution of the two branches of the W-Net and to understand how the deep network learns the ocean features from SST and SSH, we show the intermediate predictions from both the branches and the final prediction for a test image corresponding to July 8, 2016, in Fig. 5. Specifically, we plot the four-channel output from the final upconvolution block and the segmentation based on these four intermediate channels on both the SST and SSH sides separately [see Fig. 5(a) and (b)]. These layers show that both SST and SSH sides try to learn the features independently. The SSH side of the W-Net successfully captures the Gulf Stream meanders and eddies but estimates slightly more area for all features than the ground truth. The SST side of the W-Net does not capture the meanders and eddies as accurately as the SSH side; however, it helps in adjusting the size of the SSH-based predictions such that the final segmentation is correct. The final segmentation corresponds to the four-channel output obtained after applying 1×1 convolution on the intermediate outputs from both sides followed by channel-wise addition [see Fig. 5(c)]. This operation results in correcting the four

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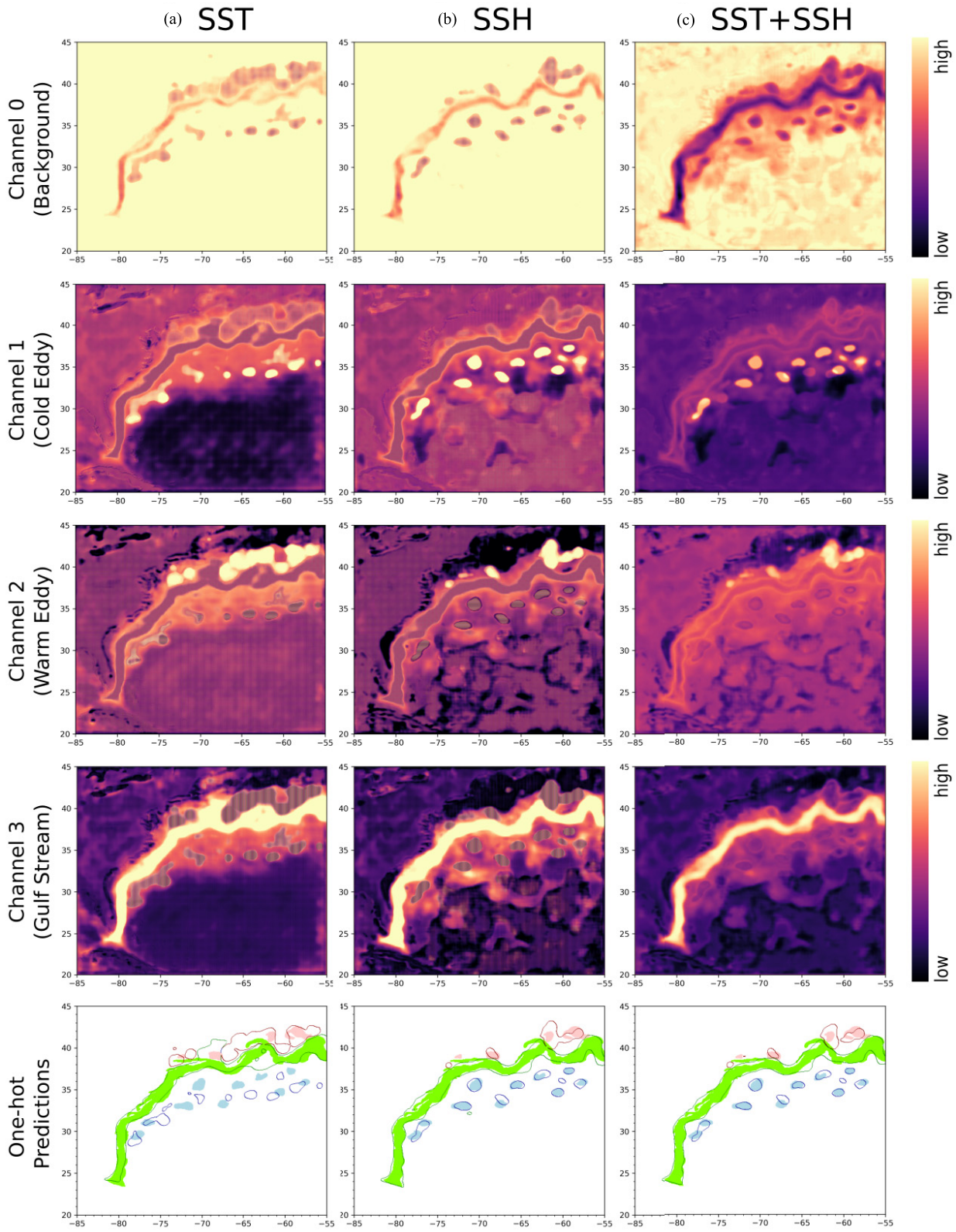


Fig. 5. Intermediate prediction analysis. (a) Four-channel and one-hot predictions from the SST part of the W-Net, after the final SST side upconvolution block (decoder), before the 1×1 convolution. (b) Same as (a) but for the SSH part of the W-Net. (c) Same as (a) but after the 1×1 convolution and channel-wise addition before the final prediction. A max-pooling layer is applied to the four-channel output to obtain the one-hot representation with the four segmentation labels. All intermediate layers are shown as color images with a low-to-high color map with the same limits.

channels, especially Channel 0 (Background), such that the final Gulf Stream and Eddy detection Dice coefficient is improved. In each column, the last row shows the segmentation obtained by using the four-channel intermediate outputs shown in that column to understand the impact of this operation. We see that the combined channel (Column C) is closer to

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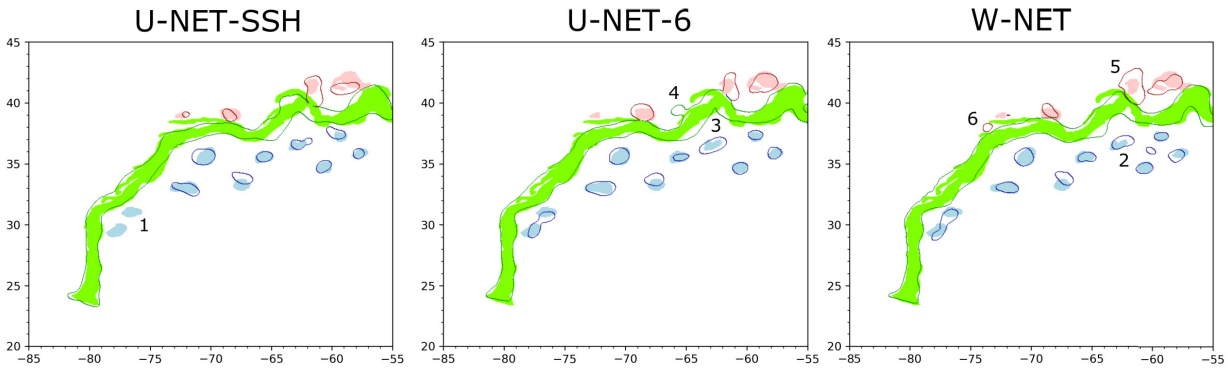


Fig. 6. Comparing predictions of U-Net-SSH, U-Net-6, and W-Net for July 08, 2016: The solid green, blue, and red segments correspond to the ground truth of gulf stream, cold eddies, and warm eddies. The hollow segments are the predictions by the three different networks. Comments about the locations marked with numbers are provided in Section IV-B.

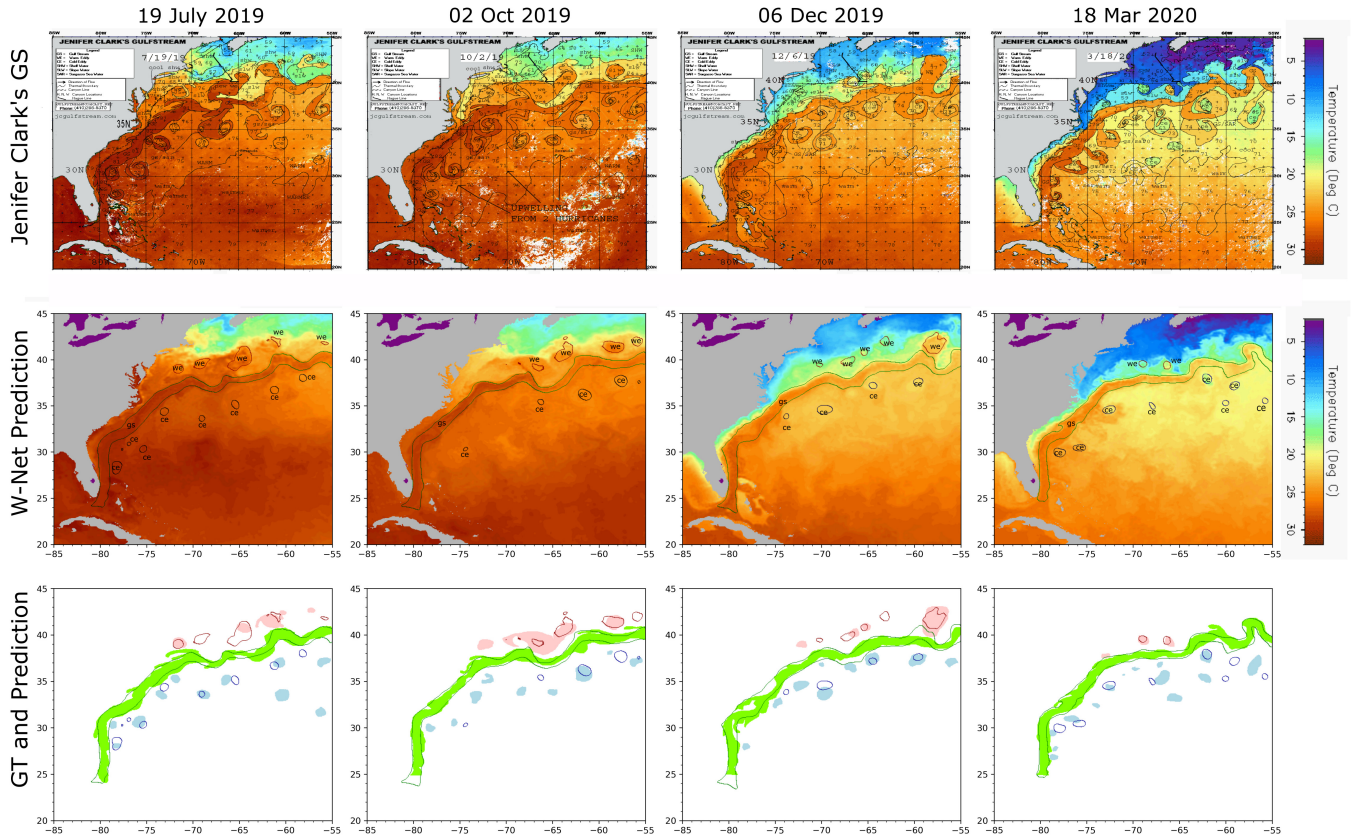


Fig. 7. W-Net predictions on 2019 and 2020 data: top row is the product from Jenifer Clark's Gulf Stream, middle row is the W-Net predictions overlaid on a background of the OSTIA SST Product, and bottom row is the ground truth extracted from top row (in solid colors) overlaid with the W-Net Predictions in the middle row. Green, blue, and red correspond to the gulf stream, cold eddies, and warm eddies segments.

the ground truth than those from the SST side (Column A) and SSH side (Column B) alone. Visually, the SSH part of the network seems to produce an intermediate segmentation very close to the final output. Hence, to further analyze the contribution of SSH for the final segmentation, Fig. 6 shows the output from U-Net-SSH, U-Net-6, and W-Net for July 8, 2016. The U-Net SSH does not capture the southwest cold eddies (marked by numeral 1), whereas the other two capture these. While the human analyst segmented two separate eddies, the deep network produced a dumbbell-shaped feature, which is acceptable. In the area marked by numeral 2, we see that W-Net predicts an extra eddy that is not present in the ground truth segmentation. Overall for the cold eddies, the prediction by a network that uses both SSH and SST is required for more accurate segmentation. Among U-Net-6 and W-Net, the segmentation of W-Net has a higher Dice coefficient (see Table I). Visually, the segmentation of July 8, 2016, qualitatively confirms this quantitative metric. For Gulf Stream, the U-Net-6 and W-Net have the same Dice coefficient, which is about 6% points higher than U-Net-SSH. However, the U-Net-6 produces a nonsmooth segmentation and often predicts artifacts (areas marked 3 and 4). On the other hand,

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the W-Net produces a smoother segmentation of the Gulf Stream, which visually looks closer to that produced by a human analyst. Sometimes, the W-Net predicts more area for warm eddies than the ground truth (areas marked 5 and 6); however, it has the best Dice coefficient on average. U-Net SSH's dice coefficient is about 20% points lower than U-Net-6 and 24% points lower than W-Net, indicating that SST information is required for improving the warm eddy detection.

Overall, taking a holistic view of the simultaneous segmentation task, the prediction from W-Net is the best both qualitatively and quantitatively. Furthermore, we see that the deep network learns complementary information from both SST and SSH. Two separate encoder–decoder networks for SST and SSH help achieve the highest accuracy for the final predictions. A comparison between the predictions of U-Net-SSH and the intermediate SSH side prediction of W-Net shows that the channel-wise addition to the intermediate layer is indeed affecting the network weights and results in an improved segmentation for W-Net.

Several expert systems and automated algorithms have tried unsuccessfully to predict the meandering Gulf Stream from either SST or SSH alone. So far, no system has used both SST and SSH with a deep convolutional network (Section I-A) for detecting the Gulf Stream. From a dynamics perspective, the convolutional filters automatically learn the multiscale gradients of SSH (through the several downconvolutional and upconvolutional blocks), representing the geostrophic currents in the region. Indeed, these currents and eddies are the signatures of the ocean features that we are trying to predict. As such, the deep network can successfully identify the segmentation's dynamical properties automatically.

C. Prediction for 2019 and 2020

The network we developed in the current work uses only data between 2014 and 2018 for training, validation, and testing purposes. Using the trained W-Net, we make predictions for 2019 and 2020, for which we did not do the ground truth generation from the Jenifer Clark maps for use in training. For these years, we simulate the use case of applying the trained network as an automated detection tool in *hindcast* mode. Fig. 7 shows four representative weeks for which Jenifer Clark's gulf stream product, the W-Net predictions, and the two overlaids on each other. A visual evaluation shows that qualitatively the network maintains the overall performance reported above. The gulf stream is detected very well, followed by warm eddies and cold eddies. To further improve the accuracy of eddies, more ground truth data should be fed into the model during training and higher resolution SSH data should be utilized. It can be seen that eddy detection struggles for October 2, 2019, likely due to the change in the ocean state from two hurricanes (marked in the Jenifer Clark image). Since only four ground truth images were generated for comparison purpose, we do not report the average accuracy metrics for 2019–2020. Most importantly, these results confirm that such a tool can be deployed in operation and utilized for various ocean and climate science applications, either as a standalone tool or to help the human analyst in a human-in-the-loop solution. Such applications include real-time identification of mesoscale features in the global ocean, automated synoptic feature-oriented ocean forecasting, automated Gulf Stream Ring census for climate change studies, and ocean biological applications to study the productivity changes.

V. Conclusion

We have developed and demonstrated a novel deep learning architecture, which we call a W-Net, to perform automated identification of gulf stream and rings (warm and cold eddies) from concurrent sea surface temperature and sea surface height images. This article is the first application of deep learning for semantic segmentation of ocean features from SST and SSH. The W-Net architecture is inspired by the popular U-Net used for biomedical image segmentation. Notably, the W-Net has two encoder–decoder U-Net-like branches: one corresponding to SST as input and the other to SSH as input. The use of SST and SSH on separate encoder–decoder networks that are finally combined was shown to perform the best segmentation. The W-Net is trained from a 5.5-km resolution SST and 27-km resolution SSH dataset. Overall, we achieve an average of 76% accuracy in detecting the warm eddies, 71.5% in detecting cold eddies, and a Dice coefficient of 82.7% for the Gulf Stream. The mean centerline path length error of the detected Gulf Stream is about 4.39% of ground truth's centerline path length. Some of the deficiencies in the detection could be attributed to the coarser resolution of SSH (27 km) and SST (5.5 km) used for the network compared to the finer resolution (1.1 km) used in the manual annotation (ground truth).

Overall, the detection accuracy is good, and this tool can serve as an input to several operational ocean forecast centers, climate scientists, and conservationists. In the future, we plan to introduce a recurrent architecture where information of past weeks' detection is utilized to improve the automated detection accuracy. Furthermore, a human-in-the-loop correction scheme can also be developed by leveraging our W-Net architecture.

Acknowledgment

The authors would like to thank the Ministry of Education, Government of India, for the graduate scholarship support to D. Lambhate and R. Sharma. They would also like to thank S. Sharma and Aniket for the help with ground truth extraction from Jenifer Clark Charts.

D. Lambhate wrote the codes, ran the experiments, produced the figures, and contributed to writing. R. Sharma developed the first version of the W-Net. J. Clark provided the manual annotation charts. A. Gangopadhyay and D. Subramani formulated the problem statement. D. Subramani designed the W-Net and variants, evaluation metrics, analyzed the outputs, debugged the software, wrote this article, and secured funding.

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Devyani Lambhate received the B.Tech. degree in computer science from the Institute of Engineering and Technology, DAVV, Indore, Madhya Pradesh, India, in 2019. She is a Graduate Student with the Department of Computational and Data Sciences, Indian Institute of Science, Bengaluru, India.

Her research interests include deep learning, computer vision, and digital image processing.



Raghav Sharma received the M.Tech. degree in computational and data sciences from the Indian Institute of Science, Bengaluru, India, in 2020.

His research interests include deep learning, computer vision, and digital image processing.



Jenifer Clark received the M.S. degree in oceanography from Johns Hopkins University, Baltimore, MD, USA, in 1975.

Affectionately known as The Gulfstream Lady, she has spent the last four decades studying, analyzing, and forecasting the World's Ocean Currents. She worked with the National Oceanic and Atmospheric Administration (NOAA), Silver Springs, MD, USA, as an Expert in the Oceanographic Features Analysis of the Gulf Stream. Her work led to numerous applications that contributed significantly to improved commercial fishing, more efficient marine transportation and commerce, and improvements in navigational safety.



Avijit Gangopadhyay received the Ph.D. degree in ocean engineering from the University of Rhode Island, Kingston, RI, USA, in 1990.

He joined UMass Dartmouth, Dartmouth, MA, USA, in 1997. He was a Research Associate with Harvard University, Cambridge, MA, USA, and a Scientist with the Jet Propulsion Laboratory, La Canada, CA, USA. He has held multiple visiting, distinguished, and honorary professorships with Harvard, the Institute of Oceanography of the University of São Paulo, Paulo, Brazil, the Indian Institute of Science, Bengaluru, India, IIT Kharagpur, Kharagpur, West Bengal, India, and IIT Bhubaneswar, Argul, Odisha, India. He is a Professor with the School of Marine Science and Technology, UMass Dartmouth, where he was the Interim Dean and an Associate Dean for seven years. His research interests include dynamics of the western boundary currents, operational ocean modeling and data assimilation, basin-scale climate-related modeling, and multiscale multidisciplinary data-model synthesis.

Dr. Gangopadhyay is the recipient of many honors, including the Chair Professorship (Ministry of Earth Sciences, India) at IIT Kharagpur, US Fulbright-Nehru Scholar in Teaching and Research (US), and Distinguished Professorships at IITKGP and IITBBS.



Deepak Subramani (Member, IEEE) received the Ph.D. degree in mechanical engineering and computation from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 2018.

He is an Assistant Professor with the Department of Computational and Data Sciences, Indian Institute of Science, Bengaluru. His research interests are in the use of machine learning and artificial intelligence in geosciences, data assimilation, autonomous routing of marine vehicles, and data-driven application systems.

Dr. Subramani is the recipient of many awards, including the Arcot Ramachandran Young Investigator Award and the de Florez Graduate Science Award.

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